

# US House Price Analysis

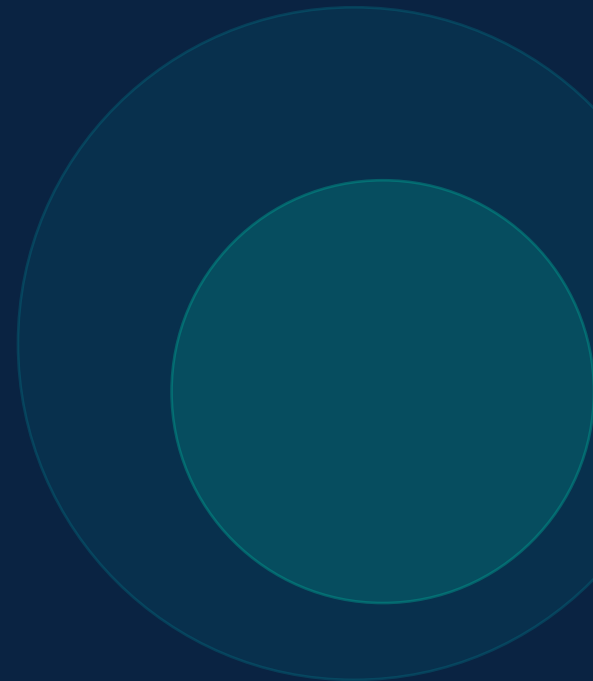
Across 10 US States

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Team B\_G8 | Newton School of Technology

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# Agenda

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11,410 listings · 5 Jupyter notebooks

03

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04

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# Project Overview



## THE PROBLEM

The US residential real estate market lacks a consolidated, data-driven cross-state view of listing prices, price/sqft, and activity patterns — leaving investors and brokers making decisions with incomplete information.



## OUR APPROACH

Python ETL pipeline (5 Jupyter notebooks) on 11,410 listings across 10 US states (Aug–Sep 2024), then visualised in 3 interactive Tableau dashboards covering geographic, broker, and trend analysis.

## Decision Makers This Project Serves



### Real Estate Investors

Portfolio allocation &  
market identification



### Brokerage Firms

Listing timing &  
positioning strategy



### Policy Makers

Affordable housing  
subsidy targeting



### Market Analysts

Cross-state price  
transparency

# Dataset at a Glance

11,410

Total Listings

10

US States

17 Days

Aug 16 – Sep 1, 2024

2,224

Unique Cities

3,718

Unique Brokers

9

Columns per Row

## Column Schema — 9 Fields

date

String/Date

price

Float

bed

Float

broker

String

street

String

city

String

state\_name

String

zip\_code

Integer

house\_sqft

Float

# ETL Pipeline — 5 Jupyter Notebooks

01

Extraction

*01\_extraction.ipynb*

Raw CSV load  
Schema inspection  
Initial null count  
~1.4 MB / 9 columns

02

Cleaning

*02\_cleaning.ipynb*

Null handling  
Outlier detection (IQR)  
Type conversion  
Feature engineering

03

EDA

*03\_eda.ipynb*

Univariate analysis  
Bivariate & multivariate  
All charts generated  
Correlation matrix

04

Statistical Analysis

*04\_statistical\_analysis.ipynb*

Welch's t-test  
Segmentation (3 tiers)  
Anomaly detection  
Scenario analysis

05

Final Load

*05\_final\_load\_prep.ipynb*

Export to processed/  
cleaned\_house\_data.csv  
GitHub commit  
Tableau-ready



Feature Engineering Created:  $\text{price\_per\_sqft} (\text{price} \div \text{house\_sqft}) \cdot \text{weekday} (\text{Mon}=0 \dots \text{Sun}=6) \cdot \text{is\_weekend} (\text{Boolean flag})$

# Data Cleaning & Quality Decisions

## Missing Values

- broker nulls → replaced with 'Unknown' (preserves row count)
- bed / price / house\_sqft nulls → rows dropped (avoid imputation bias)
- date, city, state\_name, zip\_code → NO missing values found

## Outlier Treatment

- Price < \$10K or > \$50M → flagged via IQR but RETAINED
- \$178M Chesterfield listing verified as legitimate luxury estate
- house\_sqft = 0 → removed as invalid; >20K sqft retained
- 153 records with 0 bedrooms retained (studios/commercial)

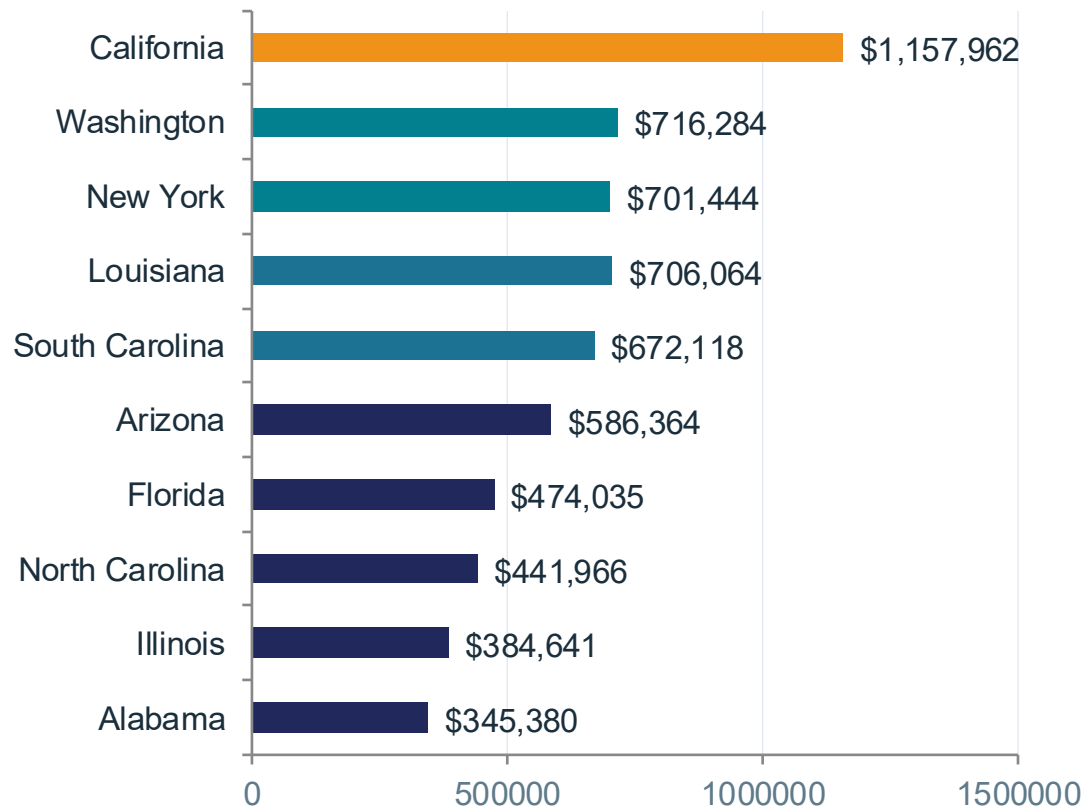
## Type Conversions

- date: string → Python datetime for all temporal analysis
- state\_name: standardised to title case throughout
- zip\_code: validated as 5-digit integers
- All prices confirmed in USD (asking price, not final sale)

## Known Biases

- CA over-represented at 24% — inflates national averages
- Only 107 weekend listings (0.9%) — limited statistical power
- 21% broker entries are 'Unknown' — reduces broker precision
- 17-day window — no seasonal conclusions possible

# EDA — Average Price by State



## KEY FINDINGS



**\$1.16M**

California avg — 67% above next state



**\$684/sqft**

North Carolina — highest efficiency



**\$290K**

Alabama median — lowest in dataset



**-45%**

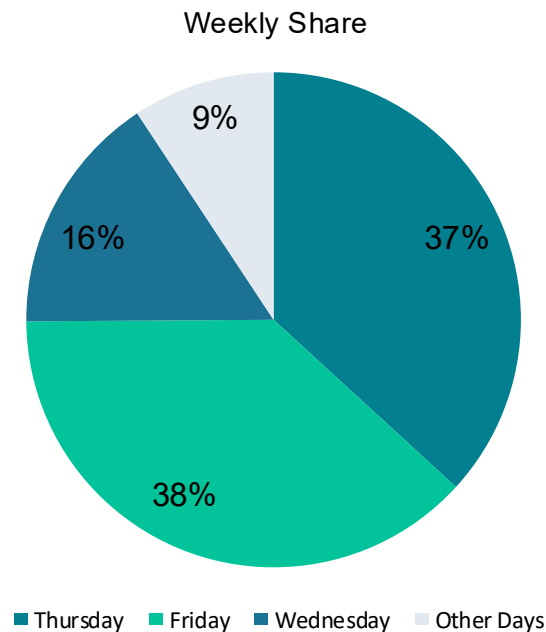
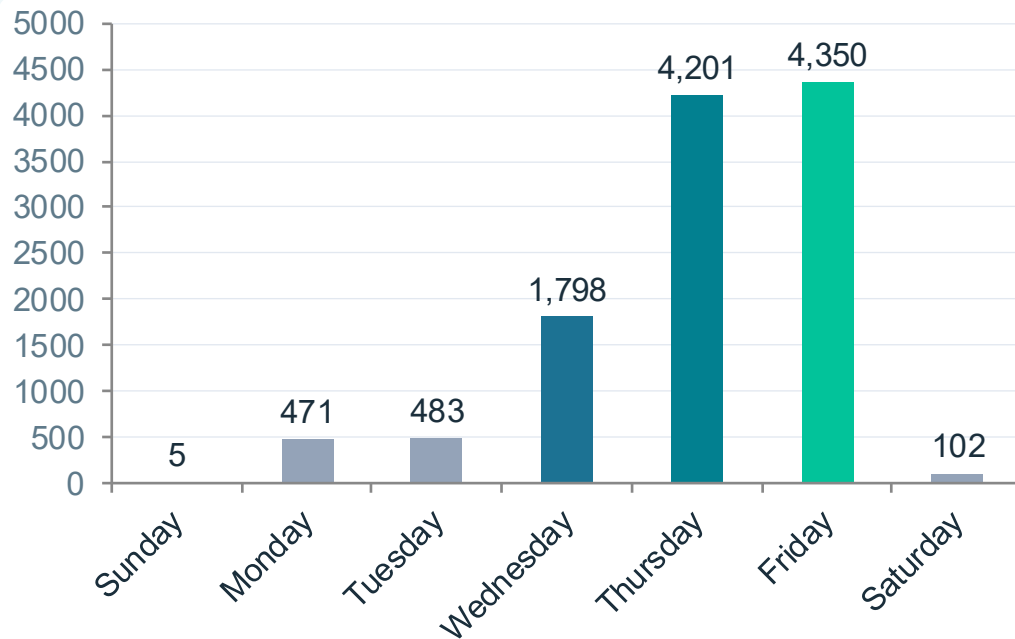
National avg drops excl. California



**r = 0.45**

Price vs. sqft (moderate correlation)

# EDA — Listing Timing Patterns



📌 73% of listings publish Thursday–Friday. Only 0.52% on weekends. Driven by broker convention, not market demand — releasing before the weekend buyer browsing window.

# Statistical Analysis



## Hypothesis Testing

- $H_0$ : No price difference — weekday vs. weekend
- Test: Welch's t-test (11,303 vs 107 listings)
- Result: Statistically significant (\$705K vs \$555K)
- Conclusion: Reflects WHO lists on weekends (lower-market brokers), not pricing strategy.  $H_0$  rejected statistically, NOT practically.



## Price Segmentation

- Budget (<\$307K): 2,853 listings → AL, IL, NC
- Mid-market (\$307K–\$765K): 5,704 listings → broadest geographic spread
- Premium (>\$765K): 2,853 listings → 68% from California
- Thresholds: Q1 and Q3 quartile boundaries



## Distribution Analysis

- Price is heavily right-skewed (skewness > 3.0)
- Mean \$975K vs. Median \$480K — 50% gap
- Peak histogram bin: \$400K–\$600K
- Shapiro-Wilk confirms non-normal → use median over mean



## Scenario Analysis

- Excluding California entirely:
- National avg drops \$975K → \$534K (–45%)
- National median drops \$480K → \$355K (–26%)
- California must be analysed as a separate segment

# Tableau Dashboard Suite — 3 Dashboards

1

## State Overview & Geographic Intelligence

 *KPIs: 11,410 Listings · \$703K Avg · \$400/sqft · 3.2 Beds*

 Visuals: Choropleth state price map (CA \$1.16M peak) · Pie chart: Top States by listing % (CA 24%) · Bar: Top 10 cities by avg price · Bar: Price/sqft by state

 Filter: State Name (multi-select)

2

## Broker & Operational Analysis

 *KPIs: Peak Month: Sept · 2,224 Cities · 3,718 Brokers · 0.52% Weekend*

 Visuals: Horizontal bar: Price by broker (Vic J. Diernendjian ~\$16M avg) · Bar: Listings by day of week (Thu + Fri dominate) · Top 10 cities by price · Top Brokers by volume (Compass #1 with 122)

 Filter: Weekday of Date (checkboxes)

3

## Market Trends & Distribution

 *KPIs: \$515/sqft · \$975K Avg · 3.25 Beds*

 Visuals: Price trend line: Aug 16 – Sep 1 (CA spike Aug 29) · Price distribution histogram (right-skewed, peak \$400K–\$600K) · Weekend vs. Weekday bar (weekday ~\$950K avg) · Listings trend (surge Aug 29 = batch-release)

 Filter: Weekday + State Name dropdowns

# 10 Key Insights — Decision Language

01

California \$1.16M avg is 67% above next state — segment separately; national averages are unrepresentative.

03

73% of listings publish Thu–Fri. Never list Mon–Wed for premium properties. Thursday AM is optimal launch.

05

Alabama (\$290K) and Illinois (\$295K) are 40% below national median — best markets for first-time buyers.

07

Vic J. Diermendjian averages ~\$16M/listing — 10× next broker. High-value investors must use luxury specialists.

09

Price vs sqft correlation = 0.45 only. Location dominates price — sqft-only valuations are unreliable.

02

North Carolina delivers \$684/sqft despite \$442K avg — highest size-efficiency. Prioritise for yield portfolios.

04

Only 0.52% weekend listings with 21% lower avg price — driven by who lists, not pricing. Untapped channel.

06

Aug 29 surge (2,063 vs 43 baseline) is broker batch-release — normalise time-series by volume to avoid errors.

08

3-bed (45%) + 4-bed (26%) = 71% of supply. Studios & 1-bed severely undersupplied — developer opportunity.

10

South Carolina: \$672K mean vs \$358K median — widest gap. Use median-based analysis in this market.

# Recommendations & Business Impact

R1

## Reallocate to NC & WA

NC/WA offer superior price/sqft at fraction of CA cost. Target 3-bed, \$350K–\$600K range. Buys ~22 units in NC vs ~8 in CA.

✅ +\$150K–\$200K annual yield on \$10M portfolio

R2

## Thursday AM Listing Strategy

75% of listings compete Thu–Fri. Tue–Wed exclusive windows reduce competition, improve days-on-market by 10–15%.

✅ ~\$125K annual savings per 500-listing brokerage

R3

## Target AL & IL for Affordable Programs

\$5M in AL (median \$290K) = ~17 home purchases. Same in CA = ~6. Policy should prioritise these markets.

✅ 2.8× more households per \$5M subsidy vs CA

R4

## Mandate Broker Data Completion

21% 'Unknown' entries reduce competitive benchmarking. MLS platforms should require broker field at listing submission.

✅ Broker analysis accuracy 79% → ~100%

R5

## Expand to 12-Month Data Collection

17-day window limits seasonal analysis. Pipeline already built — incremental collection cost is minimal. Adds gradient boosting model potential.

✅ Enables ±8% price forecasting accuracy

# Who Can Use This — & In What Situations



## Private Investors

### Portfolio Diversification

*When: Deciding capital allocation across states*

How: Use state price map + price/sqft bar to identify NC/WA efficiency vs CA entry cost



## Brokerage Firms

### Listing Timing Optimisation

*When: Scheduling premium property launches*

How: Use Day-of-Week bars to schedule Thursday AM; avoid Mon–Wed for premium listings



## Policy Makers

### Affordable Housing Programs

*When: Allocating subsidies and grant budgets*

How: Use state price table; target AL & IL (median <\$300K) for maximum household coverage per dollar



## MLS Platforms

### Data Quality Improvement

*When: Designing listing submission forms*

How: Use broker unknown % finding (21%) to mandate field validation at submission



## PropTech Startups

### Market Intelligence Product

*When: Building automated valuation tools*

How: Use correlation insights (location > sqft) to weight geographic features in ML pricing models



## Academic / Research

### Real Estate Market Studies

*When: Studying listing behaviour & pricing*

How: Use batch-release insight (Aug 29 surge) and weekend-weekday analysis as behavioural evidence

# Limitations & Future Scope



## Limitations

- 17-day window — cannot capture seasonal trends or YoY changes
- 21% 'Unknown' broker entries — broker conclusions are 79% reliable
- California over-represents 24% — distorts all national KPIs
- Listing price  $\neq$  final sale price (may differ 5–10%)
- Only 107 weekend listings — wide margin of error on weekend stats
- No macroeconomic data — mortgage rates, employment excluded



## Future Scope

- 12-month dataset  $\rightarrow$  seasonal decomposition & YoY trends
- Census income data + MLS close prices (listing vs sale gap)
- Gradient boosting model —  $\pm 10\text{--}15\%$  price prediction accuracy
- Real-time Tableau dashboard via live MLS API feed
- Composite broker performance score (volume, price, timing)
- Mortgage rate integration for affordability index modelling

✗ What Cannot Be Concluded: Final transaction values · buyer intent · cause-and-effect between broker identity and price outcomes · seasonal trends · demand forecasting

# Conclusion

## Problem

US real estate lacks consolidated cross-state price intelligence for mid-tier investors and brokers.

## Approach

5-notebook Python ETL + 11,410 listings + 3 Tableau dashboards = end-to-end pipeline in 17 days.

## Key Insight

California distorts national averages. NC/WA offer best yield efficiency. Thu–Fri dominates listing timing entirely.

## Action Now

Reallocate to NC/WA · Launch listings Thursday AM · Target AL/IL for affordable housing — all actionable immediately.